

This assignment will not be graded and is only for practice.

Key Points: 1**A matrix is in row echelon form if :**

- The first non-zero element (the leading entry) in a row is 1.
- The column containing the leading 1 of a row is to the right of the column containing the leading 1 of the row above it.
- Any non-zero rows are always above rows with all zeros.

A matrix is in reduced row echelon form if :

- The first non-zero element in the first row (the leading entry) is the number 1.
- All subsequent non-zero rows must also have their leading entries (i.e. first non-zero entries) as 1 and they should appear to the right of the leading entry in the previous row.
- The leading entry in each row must be the only non-zero number in its column.
- Any non-zero rows are always above rows with all zeros.

Note: Any matrix which is in reduced row echelon form is also in row echelon form.

- 1) Choose the correct options for the matrix $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ **1 point**

[Hint: Find out the leading entries for each rows and observe whether the matrix is satisfying the conditions of row echelon form and reduced row echelon form.]

- ☐ A is in row echelon form but not in reduced row echelon form.
- ☐ A is in both row echelon form and reduced row echelon form.
- ☐ A is neither in row echelon form nor reduced row echelon form.

No, the answer is incorrect.

Score: 0

Accepted Answers:

A is in both row echelon form and reduced row echelon form.

- 2) Choose the correct options for the matrix $A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ **1 point**

- ☐ A is in row echelon form but not in reduced row echelon form.
- ☐ A is in both row echelon form and reduced row echelon form.
- ☐ A is neither in row echelon form nor reduced row echelon form.

No, the answer is incorrect.

Score: 0

Accepted Answers:

A is neither in row echelon form nor reduced row echelon form.

- 3) **1 point**

Choose the correct options for the matrix $F = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

- ☐ A is in row echelon form but not in reduced row echelon form.
- ☐ A is in both row echelon form and reduced row echelon form.
- ☐ A is neither in row echelon form nor reduced row echelon form.

No, the answer is incorrect.

Score: 0

Accepted Answers:

A is in row echelon form but not in reduced row echelon form.

Key Points: 2

Let $Ax = b$ be a system of linear equations and suppose A is in reduced row echelon form. Assume that for every zero row of A , the corresponding entry of b is also 0 (i.e., if the i - th row of A is 0, then so is b_i .)

- If the i - th column has the leading entry of some row, we call x_i a dependent variable.
- If the i - th column does not have the leading entry of any row, we call x_i an independent variable.

4) Consider a system of linear equations:

1 point

$$0x_1 + x_2 + 0x_3 + 0x_4 = 1$$

$$0x_1 + 0x_2 + x_3 + 0x_4 = 1$$

Choose the set of correct options.

[Hint: Recall the definitions of independent and dependent variable with respect to reduced row echelon form.]

- ☐ x_1 and x_2 are dependent variables.
- ☐ x_2 is a dependent variable.
- ☐ x_3 and x_4 are independent variables.
- ☐ x_4 is an independent variable.

No, the answer is incorrect.

Score: 0

Accepted Answers:

x_2 is a dependent variable.

x_4 is an independent variable.

5)

let $Ax = b$ be a system of linear equations, where $A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$, and $b =$

1 point

$$\begin{bmatrix} 3 \\ 2 \\ 0 \\ 0 \end{bmatrix}.$$

Choose the set of correct options.

- ☐ x_1 is a dependent variable.
- ☐ x_2 is a dependent variable.
- ☐ x_3 is a dependent variable.
- ☐ The expressions $x_1 = 3$ and $x_3 = 2 - x_4$ show the dependency of x_1 and x_3 on the elements of b and the independent variables.

No, the answer is incorrect.

Score: 0

Accepted Answers:

x_1 is a dependent variable.

x_3 is a dependent variable.

The expressions $x_1 = 3$ and $x_3 = 2 - x_4$ show the dependency of x_1 and x_3 on the elements of b and the independent variables.

Key Points: 3

The three different types of elementary row operations that can be performed on a matrix are:

- **Type 1:** Interchanging two rows.
- **Type 2:** Multiplying a row with some constant.
- **Type 3:** Adding a scalar multiple of a row to another row.

Using these row operations one can obtain row echelon form or reduced row echelon form of a given matrix.

Consider the following matrix A and answer the questions (a), (b), (c), (d), and (e).

$$\begin{bmatrix} 1 & 3 & 2 & 0 \\ 0 & 2 & 1 & 0 \\ 2 & 4 & 4 & 3 \end{bmatrix}$$

6) What will be the third row of the matrix if we perform the row operation $R_3 - 2R_1$ on **1 point** A?

- ☐ (1 1 2 3)
- ☐ (4 10 8 3)
- ☐ (0 -2 0 3)
- ☐ (0 4 4 3)

No, the answer is incorrect.

Score: 0

Accepted Answers:

(0 -2 0 3)

7) What will be the second row of the matrix if we perform the row operation $\frac{1}{2}R_2$ on A **1 point** ?

- ☐ (0 1 1 0)
- ☐ (0 4 2 0)
- ☐ (0 1 $\frac{1}{2}$ 0)
- ☐ (1 2 2 $\frac{3}{2}$)

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$(0 \quad 1 \quad \frac{1}{2} \quad 0)$$

8) If we perform the row operations mentioned in question (a) and question (b), simultaneously on A , the matrix (let it be called B) we get is:

1 point

☐
$$\begin{bmatrix} 1 & 3 & 2 & 0 \\ 0 & 1 & \frac{1}{2} & 0 \\ 0 & -2 & 0 & 3 \end{bmatrix}$$

☐
$$\begin{bmatrix} 1 & 3 & 2 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 2 & 3 \end{bmatrix}$$

☐
$$\begin{bmatrix} 1 & 3 & 2 & 0 \\ 0 & 4 & 2 & 0 \\ 0 & -2 & 0 & 3 \end{bmatrix}$$

☐
$$\begin{bmatrix} 1 & 3 & 2 & 0 \\ 0 & 1 & \frac{1}{2} & 0 \\ 0 & 4 & 4 & 3 \end{bmatrix}$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\begin{bmatrix} 1 & 3 & 2 & 0 \\ 0 & 1 & \frac{1}{2} & 0 \\ 0 & -2 & 0 & 3 \end{bmatrix}$$

9) Which row operations have to be performed on the matrix B (matrix B is as in question (c)) to make all the entries 0 in the second column, except the leading entry of the second row.

1 point

☐ $R_1 - 3R_2$ and $R_3 - 2R_2$

☐ $R_1 + 3R_2$ and $R_3 + 2R_2$

☐ $R_1 + 3R_2$ and $R_3 - 2R_2$

☐ $R_1 - 3R_2$ and $R_3 + 2R_2$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$R_1 - 3R_2 \text{ and } R_3 + 2R_2$$

10) The reduced row echelon form of A is

1 point

[Hint: Recall the definition of reduced row echelon form.]

☐
$$\begin{bmatrix} 1 & 0 & \frac{1}{2} & 0 \\ 0 & 1 & \frac{1}{2} & 0 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

☐
$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

☐
$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -\frac{3}{2} \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

☐
$$\begin{bmatrix} 1 & 0 & 0 & -\frac{3}{2} \\ 0 & 1 & 0 & -\frac{3}{2} \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\begin{bmatrix} 1 & 0 & 0 & -\frac{3}{2} \\ 0 & 1 & 0 & -\frac{3}{2} \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

Key Points: 4

Let $Ax = b$ denote the matrix representation of a system of linear equations.

The augmented matrix is denoted by $[A \mid b]$.

The matrix obtained by performing the operations that transform A to its reduced row echelon form R on the augmented matrix is denoted by $[R \mid c]$.

The solutions of $Rx = c$ are the same as the solutions of $Ax = b$.

Consider a system of linear equations

$$2x_1 + x_2 = 1$$

$$-x_1 + x_3 + x_4 = -1$$

$$x_1 + x_2 - x_3 + x_4 = 2$$

$$-x_1 + x_3 + x_4 = 1.$$

Answer the questions (a), (b) and (c) based on the above information.

11) Which of the following represents an augmented matrix of the system?

1 point

[Hint: First try to write down the coefficient matrix A and the corresponding vector b .]

☐
$$\left[\begin{array}{cccc|c} 2 & 1 & 0 & 0 & 0 \\ -1 & 0 & 1 & 1 & 0 \\ 1 & 1 & -1 & 1 & 0 \\ -1 & 0 & 1 & 1 & 0 \end{array} \right]$$

☐
$$\left[\begin{array}{cccc|c} 2 & 1 & 0 & 0 & 1 \\ -1 & 0 & 1 & 1 & -1 \\ 1 & 1 & -1 & 1 & 2 \\ -1 & 0 & 1 & 1 & 1 \end{array} \right]$$

☐
$$\left[\begin{array}{cccc|c} 2 & 1 & 0 & 0 & -1 \\ -1 & 0 & 1 & 1 & 1 \\ 1 & 1 & -1 & 1 & 2 \\ -1 & 0 & 1 & 1 & -1 \end{array} \right]$$

☐
$$\left[\begin{array}{cccc|c} 2 & 1 & 0 & 0 & 1 \\ -1 & 1 & 1 & 0 & -1 \\ 1 & 1 & -1 & 1 & 2 \\ -1 & 1 & 1 & 0 & -1 \end{array} \right]$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\left[\begin{array}{cccc|c} 2 & 1 & 0 & 0 & 1 \\ -1 & 0 & 1 & 1 & -1 \\ 1 & 1 & -1 & 1 & 2 \\ -1 & 0 & 1 & 1 & 1 \end{array} \right]$$

12) What will be the matrix $[R|c]$ obtained by performing the operations that transform A to its reduced row echelon form R on the augmented matrix?

1 point

☐
$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & -1 & 0 \\ 0 & 1 & -1 & 2 & 1 \\ 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 & 2 \end{array} \right]$$

☐
$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 2 & 1 \\ 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 2 \end{array} \right]$$

☐
$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & 2 \end{array} \right]$$

☐
$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 2 & 1 \\ 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 & 2 \end{array} \right]$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 2 & 1 \\ 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 & 2 \end{array} \right]$$

13) Choose the correct option. [Hint: Write down the system of linear equations corresponding to the reduced row echelon form of the augmented matrix.]

1 point

- ☐ The system of linear equations has an infinite number of solutions.
- ☐ The system of linear equations has a unique solution.
- ☐ The system of linear equations has no solution.
- ☐ $x_1 = x_2 = x_3 = x_4 = 1$ is a solution of the system of linear equations.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The system of linear equations has no solution.

Your score is: 0/13